Daniel Olliver

Full-Stack Developer · AI Postgraduate · Quantum-Validated Research Portfolio

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PROFILE

Full-stack developer (~4 years, MERN/Next.js) who ships production software for real businesses — including a healthcare platform for a UK provider — paired with postgraduate AI training and a self-directed research program in quantum computing and geometric machine learning, validated on real IBM quantum hardware and published openly. I bring both halves to client work: rigorous, evidence-first engineering and the delivery skills to put it in production.

EXPERIENCE

Freelance Full-Stack Developer 2024 – present

Independent · Auckland (remote, AU/NZ/UK clients)

- Designed, built and shipped a production ADHD patient-assessment platform for a UK healthcare provider (React/Node stack, clinical workflows, deployment and handover).
- Migrated a tour operator's website (backtonaturetours.co.nz) to a client-editable Webflow CMS with FareHarbor booking and a full SEO migration (preserved URLs, 301s, structured data).
- Build AI-assisted delivery pipelines (Claude-based agent workflows) for faster, test-backed full-stack development; private RAG/document-AI prototypes for professional-services use cases.

Independent Researcher — Quantum Computing & Geometric ML 2025 – 2026

Self-directed; open methods at ***danielolliver.com*** *and github.com/dwatces*

- Ran ~30 controlled experiments on lattice geometry in learning and quantum systems under a strict pre-registration-style discipline (controls, matched parameters, no result recorded before its run completed).
- Demonstrated topological order on real hardware: anyon creation and braiding statistics reproduced across **three IBM quantum processors** with error mitigation and cross-device error bars; measurement-based "magic-state" programming on a 6-qubit cluster (0.999 agreement with theory).
- Computed the Kitaev model's non-Abelian phase exactly (Chern numbers, Majorana zero modes, numerically braided exchange statistics); built symmetry-equivariant neural decoders for quantum error correction with ~4× sample-efficiency gains at matched parameters (stim / PyTorch / Qiskit).
- Shipped an interactive public demo — a true stabilizer simulation where visitors braid anyons in the browser: **danielolliver.com/anyons**.

Graduate Teaching Assistant 2024

University of Auckland

Junior Developer (intern → full-time) 2022 – 2024

3PM · Auckland

- Full-stack feature development and maintenance across React/Node products; API integrations and deployment workflows.

Developer Intern 2021

OutPace

EDUCATION

Postgraduate Certificate in Artificial Intelligence 2024

University of Auckland

BSc Computer Science 2021

Auckland University of Technology

SKILLS

Web	React, Next.js, Node.js, Express, TypeScript, MongoDB, Supabase, Stripe & API integrations, Firebase, Vercel, AWS, WordPress
AI / ML	Python, PyTorch, equivariant/geometric deep learning, RAG & LLM integration (Claude agent pipelines), rigorous experiment design
Quantum	Qiskit (real-hardware workflows, error mitigation), stim, stabilizer codes & QEC decoders, topological-order experiments